

Cloud Security Review – Week 6

Practical (NET731)

1. Overview

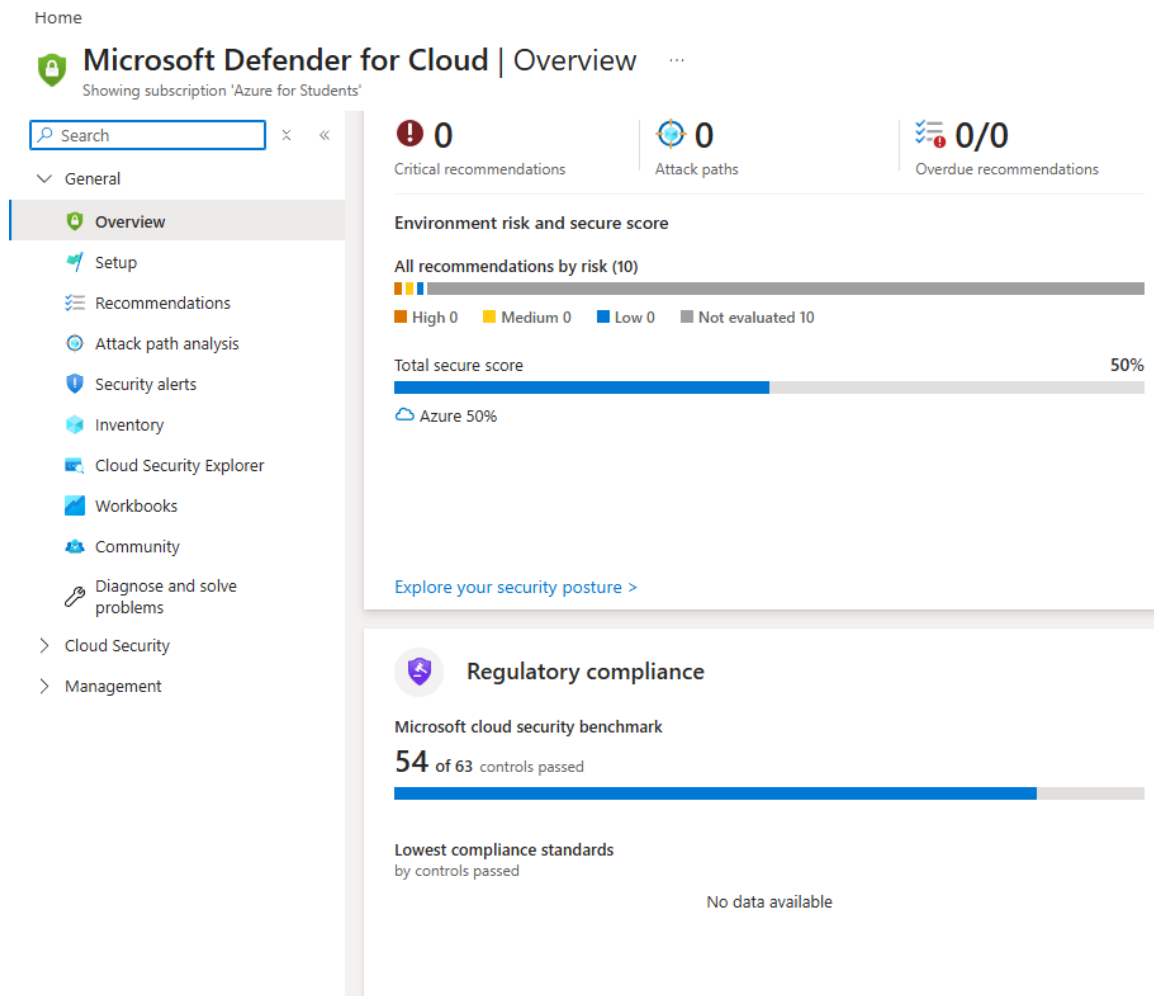
This practical focuses on reviewing and improving the security of an Azure environment using Microsoft Defender for Cloud. The task involves checking the Secure Score, identifying security risks, and applying fixes to improve the system's overall security.

1.1 Objective of the Practical

The objective of this practical is to assess the security posture of a virtual machine in Azure and apply basic security improvements. This includes identifying risks such as open ports and missing updates, and then fixing them to reduce vulnerabilities.

2. Security Posture Assessment

2.1 Secure Score



The Secure Score of the environment is 50%, which means that only some of the recommended security controls have been applied. A higher score indicates a more secure environment, while a lower score shows that there are still risks that need to be addressed.

2.2 Top 3 Security Recommendations

1. Machines should have vulnerability findings resolved

Machines should have vulnerability findings resolved ...

Open query View policy definition View recommendation for all resources

Medium Risk level week6-vm Virtual machine Unassigned Status

Description
Resolving vulnerability findings on virtual machines is a recommended step in maintaining a secure environment. These findings, identified by vulnerability assessment solutions, highlight potential weaknesses that could be exploited by malicious actors. If these vulnerabilities are not addressed, they could lead to unauthorized access, data breaches, or even system failure. Therefore, it is important to resolve these findings promptly to ensure the security and integrity of the virtual machines.

General details
Scope: Azure for Students Ticket ID: Last change date: 2025/05/05 Freshness: 12 Hours Attack Paths: 0 Risk factors: Risk: Governance: Recommendation owner: Due date:

Take action Findings Graph
Take one of the following actions in order to mitigate the threat:

- Remediate**
Review and remediate vulnerabilities discovered by the vulnerability assessment solutions.
 - Recommendation owner and set due date**
Assign owner and set due date by which recommendation should be implemented.
Assign owner & set due date
- Exempt**
Exempt the entire recommendation, or disable specific findings using disable rules. Exempted resources appear as not applicable and do not affect secure score.
Exempt Disable rule
- Workflow automation**
Set a logic app which you would like to trigger with this security recommendation.
Trigger logic app

Was this recommendation useful? Yes No

Severity: Medium

Risk: Vulnerabilities in the system can be used by attackers to gain access or damage the system.

Explanation: Fixing these vulnerabilities helps protect the system from known threats.

2. Management ports should be closed on virtual machines

Home > Microsoft Defender for Cloud | Recommendations

Management ports should be closed on your virtual machines ...

Open query View policy definition View recommendation for all resources

Take action Graph
Take one of the following actions in order to mitigate the threat:

- Remediate**
Quick fix:
Select the unhealthy resources and click "Fix" to launch "Quick fix" remediation. [Learn more >](#)
Note: After the process completes, it may take up to 24 hours until your resources move to the 'healthy resources' tab.
Fix
- Recommendation owner and set due date**
Assign owner and set due date by which recommendation should be implemented.
Assign owner & set due date
- Exempt**
Exempt the entire recommendation, or disable specific findings using disable rules. Exempted resources appear as not applicable and do not affect secure score.
- Workflow automation**
Set a logic app which you would like to trigger with this security recommendation.
Trigger logic app

Severity: Low

Risk: Open ports like SSH (22) or RDP (3389) allow attackers to attempt login attacks.

Explanation: Closing or restricting these ports improves security.

3. Machines should be configured to periodically check for missing updates

Home > Microsoft Defender for Cloud | Recommendations

Machines should be configured to periodically check for missing system updates ...

Open query View policy definition View recommendation for all resources

Take action

Graph

Take one of the following actions in order to mitigate the threat:

Remediate

Quick fix

Select the unhealthy resources and click "Fix" to launch "Quick fix" remediation. [Learn more >](#)

Note: After the process completes, it may take up to 30 min until your resources move to the 'healthy resources' tab.

Fix

To enable the periodic assessment property:

1. In the Azure portal, go to "Azure Update Manager", and select "Machines".

2. Select the relevant machines, and then select "Update settings".

3. Select "Periodic assessment" and follow the steps to complete the settings.

Recommendation owner and set due date

Assign owner and set due date by which recommendation should be implemented.

Assign owner & set due date

Exempt

Exempt the entire recommendation, or disable specific findings using disable rules. Exempted resources appear as not applicable and do not affect secure score.

Exempt

Workflow automation

Set a logic app which you would like to trigger with this security recommendation.

Trigger logic app

Prevention

Enforce remediation for future resources or Deny creation of misconfigured resources

Severity: Low

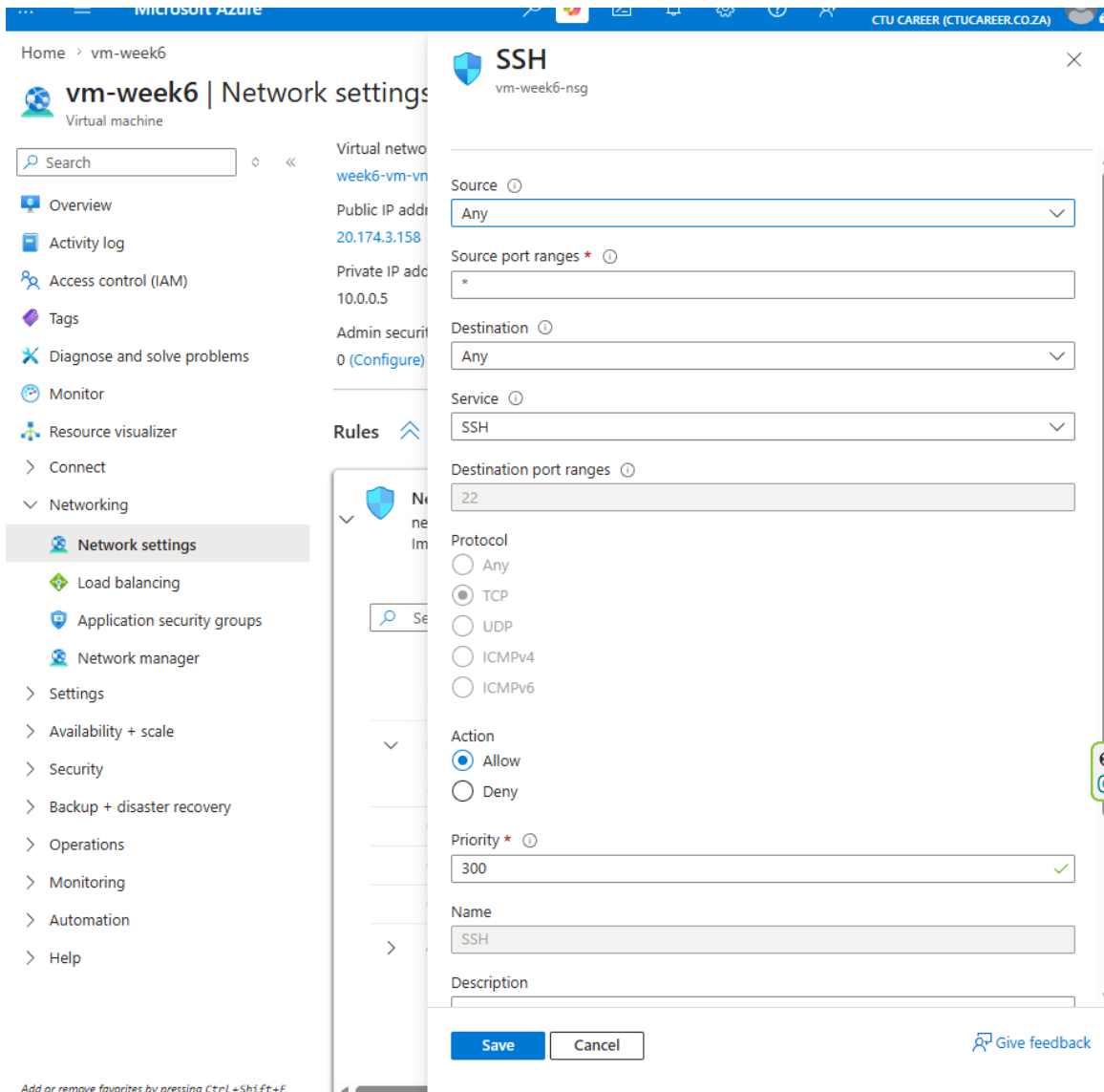
Risk: Systems without updates can be vulnerable to known security issues.

Explanation: Regular update checks help keep the system secure.

3. Remediation

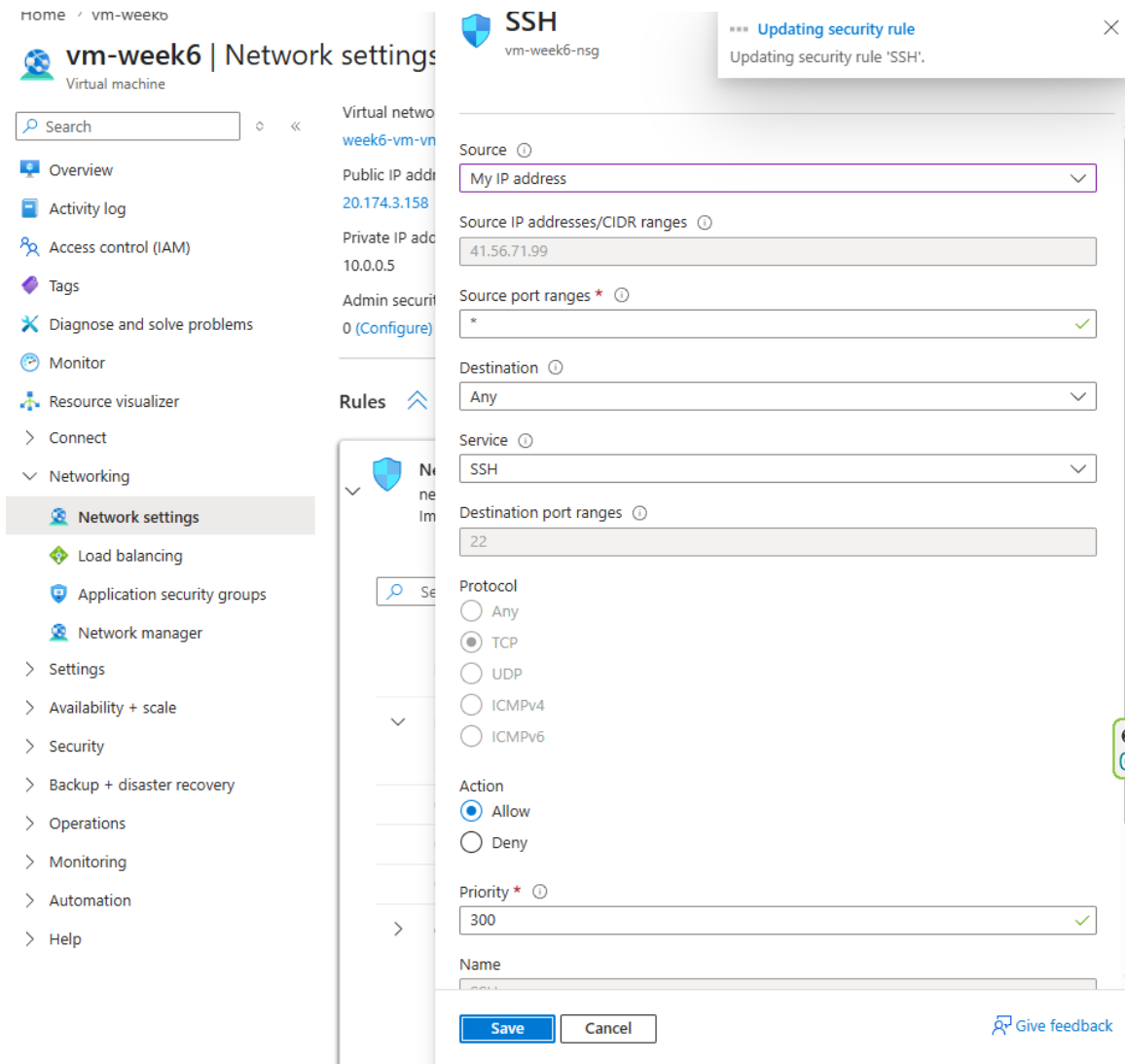
3.1 Closing Open Management Ports

Before:



The management port was open to all internet traffic, meaning anyone could attempt to connect to the virtual machine.

After:



The rule was changed to allow access only from my IP address, which limits who can connect to the virtual machine.

Explanation:

This change reduces the risk of unauthorized access by blocking unknown users from trying to log in to the system.

3.2 Enabling OS Updates

Home > vm-week6 | Updates > Updates

Change update settings

Azure Update Manager



Select the update settings that you want to change for the machines. The selected update settings will be applied to all applicable selected machines. [Learn more](#)

i Note: Patch orchestration is not applicable to Arc-enabled servers. To schedule updates on Azure machines, please change patch orchestration to "Customer Managed Schedules". This will set Patch mode to "AutomaticByPlatform" and BypassPlatformSafetyChecksOnUserSchedule to "true", which will ensure machines are patched using your configured schedules and are not autopatched. [Learn more](#)
For Hotpatch compatible Azure VMs, the patch orchestration "Customer Managed Schedules" is only allowed when Hotpatch is disabled.

Showing 1 of 1 selected resources

+ Add machine

Filter by machine name...

Resource type : All

Workloads : All

Periodic assessment : All

Hotpatch : All

Patch orchestration : All

Windows (0)

Linux (1)

Machine Name	Resource type	Periodic assessment ⓘ	Hotpatch ⓘ	Patch orchestration ⓘ
-	-	Select and apply to all ▾	Select and apply to all ▾	Select and apply to all
 vm-week6	Azure virtual machine	Enable ▾	Not available ▾ ⓘ	Image default (current)

Automatic update checks were enabled to make sure the system regularly checks for and installs updates.

Explanation:

Keeping the system updated is important because updates fix security problems and protect against new threats.

4. Conclusion

The practical successfully identified key security risks in the Azure environment and applied basic fixes to improve security. By closing open ports and enabling updates, the system is now less vulnerable to attacks. These changes help improve the overall Secure Score and security posture.